

Development and Evaluation of a Customized Wrist-Hand Orthosis using 3D Technology for a Child with Cerebral Palsy - A Case Study

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Abstract— Cerebral Palsy (CP) is a disability that affects more than 100 million children. More than 60% of these children have significant difficulties within their hand abilities, affected by involuntary movements and spasticity. So, to overcome this issue, orthoses are being employed as therapeutic intervention and can allow children with CP to have an opportunity to better use their hands. Three-dimensional (3D) technologies provide the generation of high-quality orthopedic products. Although, there are limited studies related to the evaluation of such assistive devices, especially for children. Therefore, the purpose of this research was to design, develop and evaluate a customized wrist-hand orthosis using 3D technologies for a child with CP. So, it was used a high-level 3D scanning to acquire the wrist-hand anatomy, a free software for 3D modelling and a low-cost 3D printer for manufacturing the orthosis. The Jebsen-Taylor Test of Hand Function (JTTHF) was used to evaluate unilateral hand abilities. We noticed improvements while wearing the orthosis at the affected hand, in the following subtests: writing (13 s), lifting small objects (0.9 s) and simulated feeding (69.3 s). The customized orthosis was able not only to improve some functional hand skills, but also to provide comfort, better fitting and with an appealing aesthetic design.

I. INTRODUCTION

According to the World Health Organization (WHO), over a billion people live with some kind of disability in the world. Among these amount, more than 100 million children from 0 to 14 years old are affected [1]. Cerebral Palsy (CP) is one of the most common type of childhood physical disability [2]. The overall prevalence of CP is estimated about 2.11 people per 1000 births [2].

CP is a group of permanent disorders related to movements and postures development. Such disorder causes limitations due to non-progressive disturbances, during the fetal and infant brain development [3]. In addition, CP can significantly impair a child's sensation, cognitive function, perception, communication skills, movement coordination, muscle force and behavior [3]. The most common alterations of CP are related to muscle spasticity, which is presented in

the upper limb as a flexion pattern, characterized by thumb adduction and wrist flexion [4]. This hand posture impairs most of the fine movements, resulting in disabled hands for performing daily activities, such as pinches and grasps [4]. Actually, more than 60% of children with CP have difficulties regarding their hand abilities [5].

Therapeutic interventions, in order to correct and support the hand and wrist in a functional position, are essential to allow children with CP to have an opportunity for using their hands and be able to interact with the environment [4]. Some recent studies, such as Huhges et al. [4] and Jackman et al. [5, 6], are presenting this approach, which is to provide therapeutic intervention employing functional wrist-hand orthoses for CP children.

Orthoses are considered as assistive devices and they are identified as essential goods by WHO priority assistive products list, which mentions about 50 products [7, 8, 9]. Upper limb orthoses allow disable people to live a healthy, productive, independent and dignified life, helping them to take part into social life, work and education [8, 9, 10]. Without assistive products, people who need them may suffer exclusion, isolation, poverty, and even increase the burden of morbidity and disability [7, 8].

According to WHO [8] users do wear orthoses if some of the following requirements are met: (1) they provide proper fit and alignment; (2) they do suit their needs; and (3) they are inexpensive [8]. This medical device could be customized and have all these criteria with high level by using three-dimensional (3D) technologies.

Many recent studies defend that 3D technologies are able to produce orthoses of higher quality, lighter weight, being functional, with a better fitting and more comfortable for wearing (if compared with other mass produced and ready-made orthoses) [6, 7, 8, 11, 12, 13].

The 3D technology involves 3D scanning, 3D modelling and 3D printing. Usually, 3D scanning is used to capture the geometry and measurements of the human body. 3D modelling allows to produce personalized devices, having several advantages, such as, quick producing, it offers a better fitting and more comfort for the patient [14]. Also, 3D modelling of orthosis has additional assets, since it enables to simulate its mechanical properties and pressure, which can be evaluated even before being produced [14]. 3D printing, also known as additive manufacturing (AM), enables to manufacture customized devices using a variety of materials [15]. The 3D printing has the benefits of being versatile, flexible and employing light weight materials [16].

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Healy et al. [9] mention that there are very limited studies related to the evaluation of orthotic interventions within the pediatric population. Therefore, due to this demand towards the generation of customized assistive devices, for people with disabilities, it is noticed a research opportunity to be explored. Therefore, the purpose of this research is to develop a customized functional wrist-hand orthosis using 3D technology for a child with cerebral palsy. In addition, we evaluated the functionality of the proposed hand orthosis.

II. METHODOLOGY

For the development of assistive devices, the manufacturing process consists in five stages, similar to Radoshi et al. [17], which are detailed in our methodology.

A. Volunteer background

To design a customized hand/forearm orthosis, an interview for allowing familiarity with the volunteers' needs is required. So, it involves interviewing the patient, the patient's physiotherapist, and the patient's parents as well.

B. Anatomy Acquisition - 3D geometry

Since the focus of this research is an orthosis for a child with CP, the direct scanning of the patient's anatomic region was discarded, mainly due to involuntary movements and tremor caused by the spasticity. Therefore, for the geometry acquisition we chose to make a plaster cast model (i.e. a negative impression, based on a plaster of Paris bandage), as described by Paterson et al. [18]. This process allows obtaining data, which are not burdened by errors [13, 17].

The plaster cast covers the forearm, wrist and metacarpal region. We made two plaster cast models. One from posterior view of the arm, and another one from the anterior view. After the plaster cast dried, a 3D scanning system was used for obtaining a digital 3D geometry, which was exported to the STL format file. The limb was set at a neutral position, which was determined by the volunteer's physiotherapist. We employed a hand-held system, model *REVscan* from *Creaform Ametek*®, which is based on laser scanning technology. Each cast model was placed and scanned individually on a blackboard with circular retro-reflexive targets.

C. 3D modelling and Orthosis Design

WHO [19] launched global standards, providing a guide for appropriate assistive products, including prosthetics and orthotics. The requirements suggested by the physiotherapist were similar to those specified by WHO recommendations [19]. So, the requirements design for building an orthosis are: to be light, to allow the fingers' movements, to be resistant to wrist flexion and thumb adduction, to have aesthetic look, to be soft to the skin contact, allow the patient to perform daily activities, and finally to be easy to clean.

For the preparation of the orthosis, the 3D digital geometry (STL file) was imported in the free software *MeshMixer 3.5*. This 3D modelling process includes: cleaning, smoothing, sculpting the surface, and designing the shapes (i.e. places for Velcro strap, etc.). Last, the offset was set to be 3 mm.

D. 3D Printing

In this paper, we used a low-cost 3D printer model *Prusa i3*®, based on fused-deposition modeling (FDM). *Repetier-Host 2.1.3* was used as the communication software. The material employed was a 1.75 mm white PETG filament. PETG was chosen because of its high durability, since it is not so likely of twisting, deforming or cracking. Also, it provides water resistance, ease of printability, impact resistance and slightly softer surface [20].

The *Slic3r* tool was used to slice the model into layers and to position the model into the build plate. To obtain stronger mechanical resistance, we chose to print the filaments towards the parallel lines pattern to the orthosis' length. The parameters of the printing process were set 0.3 mm layer height, printing speed 40 mm/min and infill density of 25% line's patterns.

E. Evaluation and Analysis

The final step consisted of the fitting of the orthosis within the volunteer's affected hand. While wearing the orthosis, we advise the volunteer to report any effects related to it, as skin irritation and uncomfortable pressure.

We used the *Jebsen-Taylor Test of Hand Function* (JTHF) to evaluate manual ability. The Jebsen-Taylor test was developed to evaluate unilateral hand skills involved in daily living activities [21]. JTHF have seven tasks that simulated real life activities. The tasks are: (1) writing a short sentence, (2) simulate card turning, (3) lifting small and common objects, (4) simulate feeding, (5) lifting checkers, (6) lifting large and light objects, and (7) lifting large and heavy objects [21]. The evaluation test measures the required time (in seconds) to perform each subtest.

At this research, the test was assessed with two different situations. First, without the use of the orthosis, and then using the orthosis by the affected hand. Both hands (dominant and non-dominant) are tested.

III. RESULTS AND DISCUSSION

The proposed methodology was tested with a CP child, so the results here presented are following the previous stages.

A. Volunteer background

The volunteer was a 11 years-old male child, with hemiplegic cerebral palsy diagnosis. The volunteer presents spasticity at his right hand. It is important to mention that the volunteer's dominant hand is the left hand.

B. Anatomy Acquisition - 3D geometry

Custom-made rehabilitative devices require a more complete set of body measurements and frequently includes capturing the shape and the volume of the body segment. At this research, we made a plaster mold to serve as a model [19].

We acquired both sides (posterior and anterior view from the forearm and hand) of the plaster cast, which took about 9 min. Due the spasticity, the thumb position was not in complete abduction and the wrist position was not so precisely in the neutral position (Fig. 1). The scanning process was made on the same day, after the cast model was cured. The entire scanning process demanded 19 min.

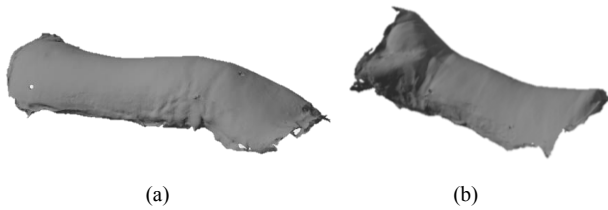


Figure 1. 3D model scanning from the plaster cast: (a) Posterior view (dorsal hand), and (b) Anterior view (palm of hand).

Dessery and Pallari [11] compared high-level with low-cost scanners. They concluded that low-cost scanners overestimate some body parts [11]. We also confirm these literature findings, as we had tested the REVscan against the Arc3D. The low-cost approaches are still losing especially towards fine details and the scale of the imaged body. For this reason, at our research - the generation of a 3D geometry of the wrist-hand of a CP child - we employed a high-level 3D scanner.

C. 3D modelling and Orthosis Design

For this research, it was designed a hole for fixing a Velcro strap and another space to support the thumb adduction. As the volunteer was a child, we also included an image as reference of his favorite superhero (as shown in Fig. 2). After finishing the modelling process, we showed it to the volunteer's physiotherapist. Then after his approval, we proceed to the 3D printing process. The entire 3D modelling process took about 8 h.

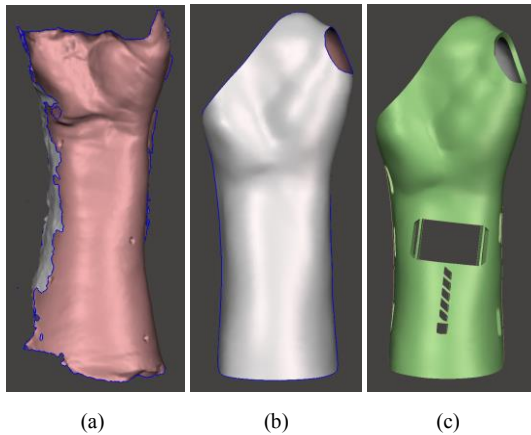


Figure 2. Results obtained from: (a) 3D scanning process, (b) 3D modelling, and (c) Design finishing.

To protect the skin, we built a glove made from neoprene®. The neoprene glove covered the forearm and extended as far as the metacarpophalangeal joints. First, the volunteer dressed the neoprene glove, and then wore the orthosis (as shown in Fig. 3). The orthosis design's decision was based on providing not only physical comfort to wear, but also social comfort acceptance.

D. 3D printing

The performed printing time was 14 h. After the printing process, the supporting material was removed, and the final orthosis was smoothed (using a P100-sanding paper). This finishing stage took 9 min. We used 124 cm³ of PETG material. Even if employing another commercial 3D printer, the printing time should be quite similar.

The generated orthosis is presented in Fig. 3. The entire manufacturing process (i.e. including 3D scanning of the plaster cast, 3D modelling and 3D printing) took about 22 h 37 min, which it is similar to the results obtained by Palousek et al. [22] (around 23 h).



Figure 3. Final orthosis: (a) Neoprene glove, and (b) the orthosis being wore by the volunteer.

E. Evaluation and Analysis

We applied the *Jebsen-Taylor Test of Hand Function* in two different situations, on the same day. First, without wearing the orthosis; and after, while wearing the orthosis. The volunteer performed each subtest with the non-dominant hand before, then with the dominant hand after. It is noticed that the volunteer is left-handed, so the right hand is his non-dominant hand. Therefore, the volunteer has only one affected hand (with spasticity), which is the right hand - his non-dominant hand.

Fig. 4 presents the results evaluation, by comparing the required time from the affected hand to perform the seven subtests, in different conditions (without and with the hand orthosis). We observed improvements in (1) writing - 13 s, (3) lifting small objects - 0.9 s and (4) simulated feeding - 69.3 s activities. In contrast, the other activities, such as (2) card turning, (5) checkers, (6) lifting light and (7) heavy objects did not reach better timing. Although, it is very important to report that without using the orthosis, the volunteer was not able to do the simulated feeding (task 4) and reached just one bean at the proposed time. However, with the orthosis, the volunteer was able to collect all the beans, allowing the child to feed himself without any additional help. The essential about this task is since it involves not only feeding, but actually holding any tool in order to carry and transfer things from one place to another, allowing more independence.

The total required time to complete all the proposed tasks without the orthosis reached 295.2 s. On the other hand, while wearing the orthosis, it was 237.2 s – implying a reduction of 58 s; while the non-affected hand had a reduction of only 10.2 s. However, the focus at this research is to test the affected hand.

Furthermore, wearing orthoses have important social benefits, apart from improving physical impairments, allowing better mobility and functionality. It helps the volunteers to remain active and productive, increasing their independence, social interaction, participation in society and providing further working opportunities [8]. About the long-term benefits from wearing the orthosis and the inclusion of more volunteers, it is still being further expanded.

The positive impact for developing custom-made assistive products goes far beyond improving health and

well-being of the individual users. For instance, there are also social-economic benefits, it helps towards shortening hospital stays, it enables employment's inclusion and working-related productivity, among other benefits [7, 8].

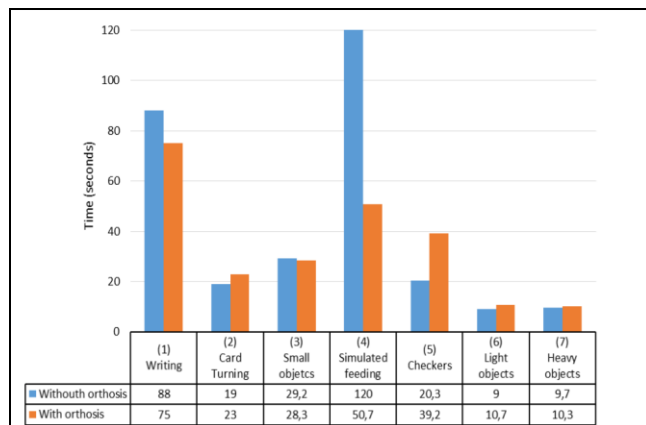


Figure 4. Results from the affected Hand by the Jebsen-Taylor Test.

IV. CONCLUSION

WHO estimates that around 35 to 40 million people worldly require prosthetics and orthotics services; but nowadays, only 10% of those in need have access to these assistive products [7]. The perspective at the next decade is that the number of people requiring prosthetics and orthotics services will rise [8]. The 3D technology has been showing great potential for the development of rehabilitative devices. Furthermore, the healthcare industry is the third largest segment of AM industry, which is growing due to the increasing demand for custom-made devices and the easy access to 3D printers [15].

In addition, since each person with disability is unique and has his/her own specific needs and remaining abilities, the hand orthosis should be customized. That is why the 3D technological approach has increasing towards the development of new orthoses, as specific requirements can modify previous designs, to make personalized and customized assistive devices. In the case study here presented, we developed and evaluated a customized wrist-hand orthosis using such technology. This orthosis provided a better fitting, comfort and improved its functional hand skills.

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